

THE CENSUS — FINDINGS

Every Boolean function on four variables, examined exactly

Reproduce: `python3 census.py` · runtime ~4 min · deterministic · exact integer arithmetic

Scope: all $2^{16} = 65,536$ functions. Not a sample. Not an estimate. The complete population.

WHY THIS EXISTS

The memo mapped the walls of P vs NP by citing theorems. This does something different: it **computes the ground truth** for an entire universe of Boolean functions, then tests the claims against it. Nothing here is asserted from memory. Every number was produced by the run.

Four variables is small enough to be exhaustive and large enough to be interesting. It is also — as the data shows twice below — sometimes *too* small, which is itself worth knowing.

1 • THE INVARIANT, VERIFIED 65,536 TIMES

Parseval: min 1.000000000 · max 1.000000000

$\sum \hat{f}(S)^2 = 1$ for every function without a single exception, to nine decimal places. Your $\sin^2 + \cos^2 = 1$, checked against the complete population rather than a demo. The conservation law is not approximately true or usually true. It is exactly true, everywhere.

2 • HUANG’S THEOREM — ZERO VIOLATIONS

In 2019, Hao Huang proved the Sensitivity Conjecture — open since 1992 — in a paper two pages long. The statement:

$s(f) \geq \sqrt{\deg(f)}$ for every Boolean function

Tested against all 65,536:

Result	Value

Functions checked	65,536
Violations	0
Slack $s - \sqrt{\deg}$	min 0.0000 · mean 1.6000 · max 2.2679
Functions achieving zero slack	104

The boundary is real and it is touched. 104 functions sit exactly on the curve — the theorem is not merely true here, it is **tight**, which means it cannot be improved without changing its form.

Why this matters for everything we discussed

Huang's proof is the invariant strategy, succeeding. He put $\pm$ signs on the edges of the hypercube, formed the matrix

$$A_1 = [[0,1],[1,0]] , A_k = [[A_{k-1}, I], [I, -A_{k-1}]]$$

and read its **eigenvalues**. That spectral invariant is the entire proof. The run rebuilds it from scratch:

Check	Result
A_4 dimension	16×16
$A^2 = 4 \cdot I$?	max deviation 0.00e+00 — exact
Eigenvalues	+2.0000 × 8 , -2.0000 × 8
All $ \lambda = \sqrt{n}$?	True
Graph consequence, 20,000 random subsets	minimum max-degree 2 $\geq \sqrt{4}$ — consistent

Every eigenvalue has magnitude exactly $\sqrt{n}$. Cauchy interlacing then forces any induced subgraph on more than half the cube to contain a vertex of degree $\geq \sqrt{n}$, and the thirty-year problem falls out.

Find the invariant, and use it to separate. That is what you kept saying. It is what worked, seven years ago, on the adjacent problem.

3 · SHANNON'S COUNTING ARGUMENT, MADE EXACT

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Measure	0	1	2	3	4
sensitivity $s(f)$	2	8	692	28,328	36,506
block sensitivity $bs(f)$	2	8	668	28,352	36,506
degree $\deg(f)$	2	8	212	12,648	52,666
decision-tree depth $D(f)$	2	8	156	8,240	57,130
certificate complexity $C(f)$	2	8	668	28,352	36,506

80.36% of all functions have full degree.

Hardness is not the exception. It is the overwhelming default — and this is exactly the *largeness* condition of Razborov–Rudich, measured instead of assumed. Easy functions are the vanishing minority. We simply cannot point at a single specific hard one and prove it.

All standard inequalities verified exhaustively: $s \leq bs \checkmark \cdot bs \leq C \checkmark \cdot C \leq D \checkmark \cdot \deg \leq D \checkmark \cdot bs \leq \deg^2 \checkmark$. Not cited — tested, 65,536 times each.

4 • TWO THINGS THE RUN TAUGHT ME

Unplanned finding: $C(f) = bs(f)$ for every single function

The distributions for certificate complexity and block sensitivity came out byte-identical, so the run checked whether they agree pointwise. They do — **all 65,536**.

General theory only guarantees $bs \leq C \leq bs^2$, and functions separating them are known. So this is not a discovery that they are equal; it is a demonstration that **four variables is too small a world to hold the counterexample**. A useful reminder about exhaustive search at small n : absence of a gap is not evidence of no gap.

Self-correction: my detector was much weaker than I claimed

I wrote that the spectral detector “works perfectly.” The run printed 74.91% accuracy directly above that sentence. So the script was changed to sweep every threshold and report the true best:

Best threshold	0.07
Best accuracy	82.09%

Trivial baseline (guess "hard" always)	80.36%
Real improvement	+1.73 points

A single spectral summary statistic is a **weak** proxy for genuine complexity. That is the honest result, and it is more informative than the flattering one: real Fourier lower bounds do not work by thresholding one number. They exploit the whole spectrum's behaviour *under random restriction* — which is why Håstad's switching lemma is a theorem and my threshold is a parlor trick.

5 · THE BARRIER, IN MEASURED QUANTITIES

Both Razborov–Rudich conditions, established numerically rather than argued:

- **Constructive** — the detector is one Walsh–Hadamard transform, 64 operations, milliseconds. Anyone can run it.
- **Large** — 80.36% of all functions carry the property.

Constructive + large = **natural**. At $n = 4$ there is no cryptography to break, so nothing stops it working. That is precisely why the barrier is *invisible at this scale and fatal at the next one*. You cannot see the wall from inside a world too small to contain a pseudorandom function.

6 · WHAT THIS DOES AND DOESN'T ESTABLISH

Does: exact ground truth for a complete function universe · exhaustive verification of a 2019 theorem and its proof object · numerical measurement of the naturalness conditions · a demonstration that single-statistic detectors are weak.

Doesn't: say anything about asymptotics. $n = 4$ is a laboratory, and §4 shows twice how a laboratory can mislead — once by hiding a real gap, once by flattering a weak detector. Everything that matters about P vs NP happens at $n \rightarrow \infty$, where exhaustion is impossible forever.

The run corrected me once, and surprised me once. That ratio is about right for an experiment worth running.